MCA FINAL PROJECT · NATURAL LANGUAGE PROCESSING

Social Engineering Message Classification using NLP

A multiclass NLP & machine-learning pipeline that detects phishing, malware, scareware, baiting and pretexting in text messages.

Presented by: Maharshi

Program: MCA — Final NLP Project



ROADMAP

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OVERVIEW

Introduction



Background

Social engineering attacks — phishing, malware, scareware, baiting and pretexting — manipulate people through deceptive text messages rather than exploiting software.



Motivation

Manual screening of such messages is slow and error-prone. Automated detection can strengthen spam filters and security-awareness tooling at scale.



Scope

Build a multiclass NLP classifier that labels a message into one of six categories using classical text-processing and machine-learning techniques.

THE CHALLENGE

Problem Statement

Existing Problem

Most spam/phishing filters perform only binary classification (safe vs. malicious) and do not distinguish between different attack types.

Project Objectives

- Build a labeled multiclass dataset
- Preprocess and vectorize the text
- Train and compare multiple ML classifiers
- Deploy the best-performing model for real-time classification



Binary-only filters

Most spam/phishing filters simply flag “safe vs. malicious” — the specific attack tactic is lost.

PRIOR WORK

Literature Review



TF-IDF + Classical ML

Naive Bayes, SVM and Logistic Regression on TF-IDF features remain a strong baseline for text-based spam and phishing detection.



Binary-Only Focus

Existing spam-detection research is largely binary (spam vs. ham), with limited focus on distinguishing specific social-engineering tactics.



Ensemble Benchmarks

Ensemble methods such as Random Forest are commonly benchmarked alongside linear models for short-text classification tasks.



This Project's Basis

Builds on these established techniques for a finer-grained, 6-class social-engineering classification problem.

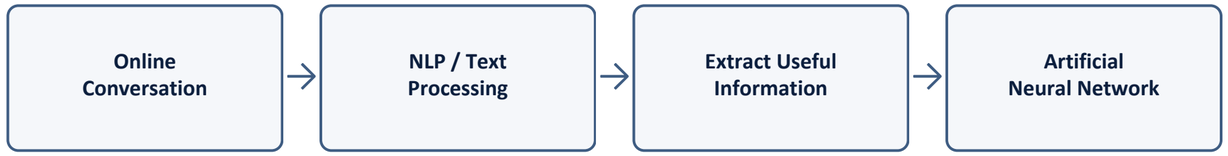
REFERENCE

Research Inspiration

Primary Paper

"SEADer: A Social Engineering Attack Detection Method Based on Natural Language Processing and Artificial Neural Networks"

M. Lansley, N. Polatidis, S. Kapetanakis, F. Mouton
Journal of Information and Telecommunication, 2020



Why This Paper Mattered

- Directly related to social engineering detection
- Uses NLP + machine learning
- Demonstrates automated attack detection
- Provides a research-based methodology
- Inspired the text-classification approach
- Highlighted feature extraction & model evaluation

WHAT'S MISSING

Research Gap



Limitation

Prior binary spam/phishing classifiers cannot separate phishing from malware, baiting, scareware or pretexting attempts.



Gap

Little existing work targets multiclass identification of specific social-engineering tactics within short text messages.



Our Solution

A TF-IDF + multiclass ML pipeline trained on a labeled corpus spanning five attack types plus benign messages.

SPECIFICATIONS

Requirement Analysis

Functional Requirements

- Load and clean the dataset
- Extract labels from raw corpus
- Preprocess text (URL/HTML removal, stopwords, lemmatization)
- Extract TF-IDF features
- Train and evaluate multiple classifiers
- Expose a predict_message() function

Non-Functional Requirements

High Accuracy

Strong classification accuracy and F1-score across all six classes.

Reproducibility

Fixed random_state ensures results are repeatable end-to-end.

Lightweight Inference

Fast enough for real-time message classification.

APPROACH

Methodology

01



Data Collection

624 labeled messages sourced from phishing_nlp_dataset.xlsx.

02



Preprocessing

Label extraction, de-duplication, text cleaning, stopword removal and lemmatization using NLTK.

03



Modeling

TF-IDF vectorization (unigrams + bigrams) feeding four candidate classifiers.

04



Evaluation

Accuracy, precision, recall and weighted F1-score on an 80/20 stratified train-test split.

DESIGN

System Architecture



Data Source

phishing_nlp_dataset.xlsx — 624 raw labeled records.



Processing

Regex label extraction, text cleaning, NLTK preprocessing, TF-IDF vectorization (1,924 features).



Model

Logistic Regression, Naive Bayes, Linear SVM and Random Forest benchmarked against each other.



Output

Predicted class via predict_message(); best model and vectorizer persisted with joblib.

PROJECT FLOW

Workflow



Problem

Identify the need for multiclass social-engineering message detection.



Literature

Review TF-IDF and classical ML approaches to text classification.



Development

Data cleaning → preprocessing → TF-IDF → model training → evaluation → best-model selection.



Conclusion

A deployable classifier achieving 88.6% accuracy with a Linear SVM.

TOOLSET

Technology Stack

Language & Environment

Python, Jupyter Notebook

Data Handling

Pandas, NumPy

NLP

NLTK (stopwords, WordNet lemmatizer), Regex

Machine Learning

scikit-learn — TF-IDF, Logistic Regression, Naive Bayes, Linear SVM, Random Forest

Visualization & Persistence

Matplotlib, joblib

BUILD

Implementation

Dataset

phishing_nlp_dataset.xlsx — 624 raw records, cleaned to 612 after removing duplicates and missing labels.

Features

TF-IDF, max 10,000 features, 1–2 grams, min_df=2, max_df=0.95 → 1,924 features.

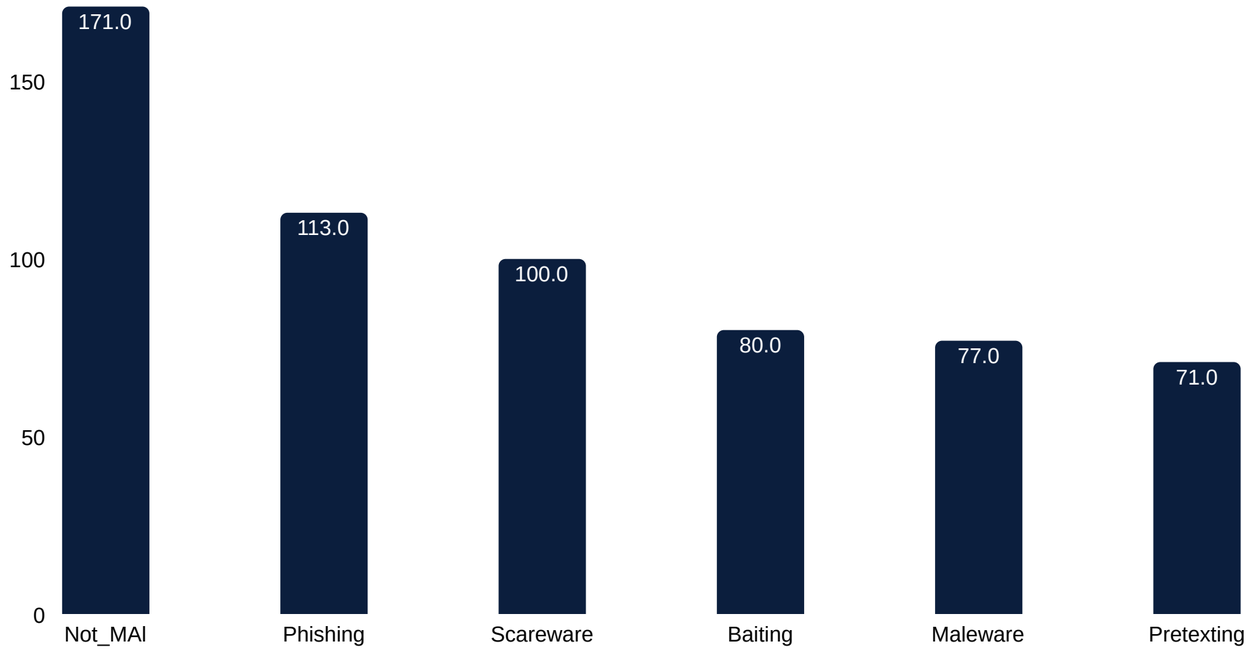
Algorithms

Logistic Regression, Multinomial Naive Bayes, Linear SVM, Random Forest (200 trees).

Split

80/20 stratified train-test → 489 training / 123 testing samples.

Class Distribution (n = 612)



FINDINGS

Results & Discussion

88.6%

Accuracy — Linear SVM
(weighted F1: 0.884)

Model	Accuracy	Precision	Recall	F1 Score
Linear SVM	**0.886**	**0.885**	**0.886**	**0.884**
Naive Bayes	0.878	0.882	0.878	0.877
Logistic Regression	0.870	0.884	0.870	0.870
Random Forest	0.870	0.873	0.870	0.867

Model ranking (F1): Linear SVM > Naive Bayes > Logistic Regression ≈ Random Forest
Most confusion occurs between semantically close classes (e.g. Phishing vs. Scareware), which share urgency-driven language.

Sample Predictions

“Your bank account has been suspended. Verify your password.” — **Phishing**

“Congratulations! You've won a free prize. Click to claim.” — **Phishing**

“Please find attached the report for tomorrow's meeting.” — **NOT-Malicious**

WHAT'S NEXT

Future Scope

Deep Learning Models



Explore transformer models (e.g. BERT) for richer contextual understanding of messages.

Dataset Expansion



Expand and balance the dataset, particularly the under-represented Malware and Pretexting classes.

Real-Time Deployment



Deploy the classifier as a real-time API or browser/email-client extension.

Explainability



Add feature importance or SHAP to highlight the persuasive keywords driving each prediction.

WRAP UP

Conclusion

Built and compared four ML models for multiclass classification of social-engineering messages. The TF-IDF + Linear SVM pipeline achieved the best result — 88.6% accuracy and 0.884 weighted F1 — across six categories.

Six-class social-engineering detector, ready for real-time inference.

References

- M. Lansley et al., “SEADer: A Social Engineering Attack Detection Method Based on NLP and ANN,” Journal of Information and Telecommunication, 2020.
- scikit-learn documentation
- NLTK documentation
- Course dataset: phishing_nlp_dataset.xlsx

Thank you.